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Laboratory Exercise: Phylogenetic Analysis Using NCBI and Galaxy

BIO 300 - Cell and Molecular Biology Laboratory

Group 3 - *Canidae*

Submitted to:

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Introduction

Understanding the evolutionary relationships among organisms is an important part of studying biology. One way scientists do this is by comparing DNA sequences to determine how closely different species are related through phylogenetic analysis. The mitochondrial cytochrome c oxidase subunit I (COI/COX1) gene was used because it is a reliable genetic marker that contains enough similarities and differences to distinguish related species (Antil, S et al 2023).

COI/COX1 nucleotide sequences of selected members of the family *Canidae* with one outgroup were obtained from the NCBI database, aligned using the Concatenate datasets tail-to-head (cat) tool in the Galaxy platform, and analyzed to generate a phylogenetic tree. The resulting tree allowed the evolutionary relationships among the selected species to be examined and compared with findings from published scientific studies.

This study addresses the biological question: What are the evolutionary relationships among the selected members of the family *Canidae* based on their mitochondrial COI/COX1 gene sequences? According to the phylogenetic analysis, species that share a more recent common ancestor are predicted to cluster together on the phylogenetic tree (Bardebelen et al 2005). On the other hand, it is expected that species with higher genetic divergence will split off into separate branches signifying a more distant evolutionary relationship. The resulting phylogenetic tree will offer a visual representation of these relationships, enabling the identification of closely related groups, the evolutionary divergence, and the assessment of the relationship between the taxonomic classification of the chosen family and the genetic variation within the COI gene.

Table of Species

Table 1. List of selected species with corresponding nucleotide accession numbers and sequence lengths.

Scientific name	Common name	NCBI	Length (bp)	Focal taxon / Outgroup
<i>Canis lupus</i>	Gray wolf	MZ099455.1	658	Focal Taxon
<i>Canis aureus</i>	Golden jackal	PP527155.1	643	Focal Taxon
<i>Canis latrans</i>	Coyote	AF028188.1	588	Focal taxon
<i>Canis simensis</i>	Ethiopian wolf	L29413.1	588	Focal taxon
<i>Lycaon pictus</i>	African wild dog	AF028195.1	588	Focal taxon
<i>Nyctereutes procyonoides</i>	Raccoon dog	AF028197.1	588	Focal taxon
<i>Felis catus</i>	Domestic cat	KF297802.1	658	Outgroup

Methods

Nucleotide sequences were obtained from the National Center for Biotechnology Information (NCBI) database by searching for each selected organism and specifying the mitochondrial cytochrome c oxidase subunit I (COI/COX1) gene region. The corresponding NCBI accession numbers and sequence lengths in base pairs (bp) were recorded for reference. After retrieving all required sequences, the downloaded files were renamed according to their respective scientific names to ensure proper identification and organization of the datasets.

The collected sequences were then uploaded into the Galaxy platform, where individual datasets were combined into a single file using the Concatenate Datasets tail-to-head (cat) tool. The concatenated dataset was then downloaded and manually edited to remove unnecessary header information, resulting in a cleaner sequence file for downstream analysis. The edited file

was re-uploaded to Galaxy and subjected to multiple sequence alignment using the MAFFT (Multiple Alignment using Fast Fourier Transform) tool to generate aligned nucleotide sequences suitable for phylogenetic analysis.

To minimize the influence of sequencing errors and improve sequence comparability, the aligned sequences were further processed for consensus analysis. The MAFFT output file was converted from FASTA format into mothur.align format to ensure uniform sequence length and maintain co-linear alignment positions before running the Consensus.seq tool. Finally, the aligned dataset generated from MAFFT was used for phylogenetic tree construction through the FastTree tool, which produced the phylogenetic tree based on the evolutionary relationships among the selected taxa.

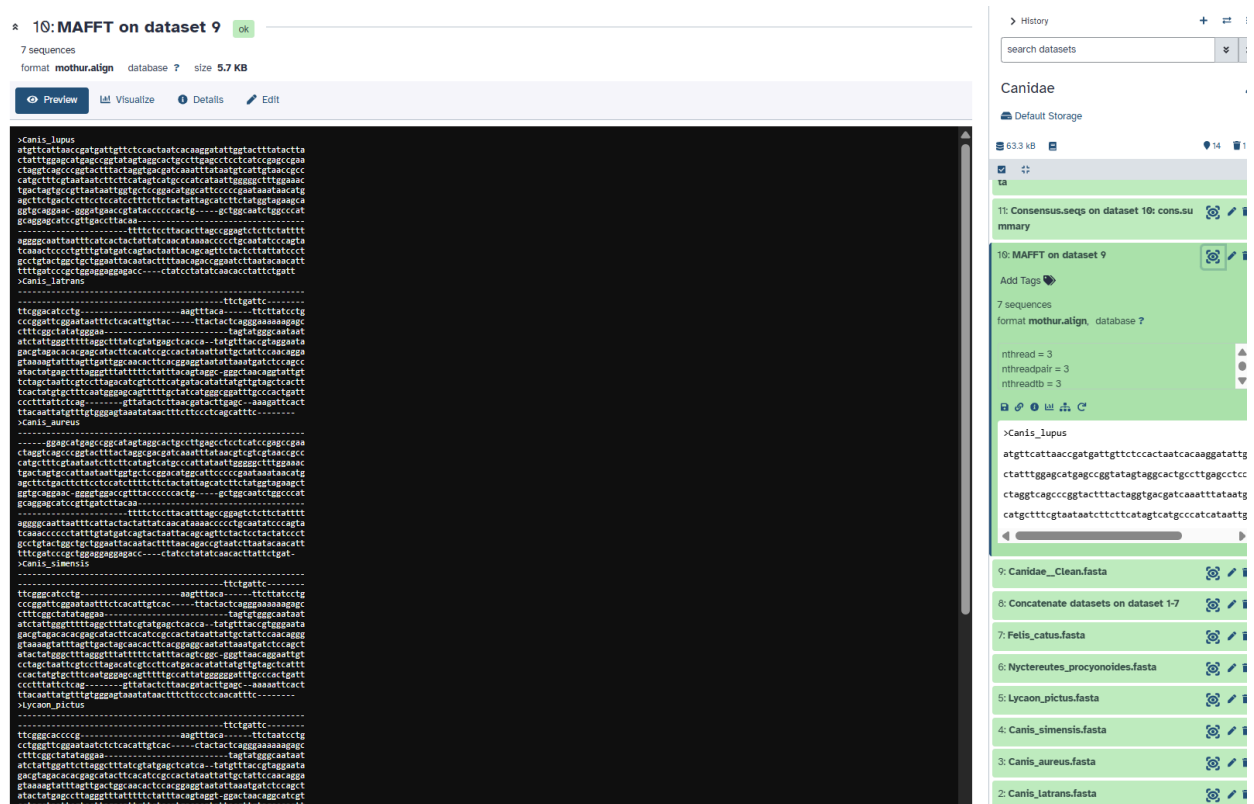


Figure 1. MAFFT alignment screenshot: Gaps were replaced with dash (-) to make every sequence the exact same length. This transforms the raw, unaligned sequences into uniform, co-linear alignment sequences

Consensus Sequence Results

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>conseq
yacyytwtatytwytrtttyggwgcagcyvrhayrnrtrrrbmcycgytvagcctyythatygcwgccgarytrnggtcarccyggkrcyytrytrggvrabracycmrathkayawy
vthryyswracyrttcycatgywyyrwrakrwttytttymyvgkyatrccyaktakdmkkrnsghwtwyggraahtkaytccaaatccangtdmsdytrakrwwyrhvcycmvga
yrtrgcvtttyccyrnakaawyarhatragygttgatataagwtykgRYYKCYBCCHCYRKsaggggtcaaaaaaggttggttaaggYTTYBWWYVVYTWBKRGVVTgtgaYYYCTR
YRGVYRRDRCDGGTagtgataRYAGSRAYDGGRYgracygthtaYSCHSYDMDRMBRSYRAYCWDRCB MAYRMAGGRGcatcmgt dgayytwacaawYTKHTMYKRSAYHTWGCHG
GVRKBTYYataTYNATWHYDGGKGRYWATHAAHTTHATYRCNMCTAKHATYRAYatRRADMCCCCNGCHAKRTSYMARKABMARAcycyytrtTYGTRWGRTCmrTMYTRATgctcct
gcatgggctaggttgccagctaattggggggYACRSMRKYHHHTYBTNYTRYSNYTRYCYHSTVYYatagaRGMWGCYRGDAWYASHAKAMWNBWDRSRGRBMGWAAYCWWAABMYHA
VVTBTYYRWYCchgcyygrrggrgdrayschaybytataycaacacytrttytgatttttcggacaccctgaagtttacattcttattctaccggattcggataatctctcacat
cgtcacctattactcagggaaaaagagcctttcggctatataggaatggtatgggcaataatctatcggttctta
```

Figure 2. Consensus sequence generated from the aligned COI sequences.

The consensus sequence was generated using the Consensus.seqs tool in Galaxy with the default majority-rule settings. The consensus sequence is 905 bp long and contains both conserved nucleotides, including A,T,C, and G. It also shows IUPAC ambiguity codes (R, Y, W, K, M, S, B, D, H, and V) representing the variable sites among the aligned taxa.

PositioninAlignment	A	T	G	C	Gap	NumberofSeqs	ConsensusBase
1	0.142857	0	0	0	0.857143	7	a
2	0	0.142857	0	0	0.857143	7	t
3	0	0	0.142857	0	0.857143	7	g
4	0	0.142857	0	0	0.857143	7	t
5	0	0.142857	0	0	0.857143	7	t
6	0	0	0	0.142857	0.857143	7	c
7	0.142857	0	0	0	0.857143	7	a
8	0	0.142857	0	0	0.857143	7	t
9	0	0.142857	0	0	0.857143	7	t
10	0.142857	0	0	0	0.857143	7	a
11	0.142857	0	0	0	0.857143	7	a
12	0	0	0	0.142857	0.857143	7	c
13	0	0	0	0.142857	0.857143	7	c
14	0	0	0.142857	0	0.857143	7	g
15	0.142857	0	0	0	0.857143	7	a
16	0	0.142857	0	0	0.857143	7	t
17	0	0	0.142857	0	0.857143	7	g
18	0.142857	0	0	0	0.857143	7	a
19	0	0.142857	0	0	0.857143	7	t

Figure 3. Partial Consensus Summary showing the nucleotide frequencies and consensus bases for aligned COI bases.

A representation of the consensus summary, which displays the frequency of each nucleotide (A, T, G, and C), gaps (represented by dash lines), and the consensus base at every position in the alignment. Standard DNA bases indicate conserved sites where most or all

sequences share the same nucleotide, while IUPAC ambiguity codes indicate variable sites where different nucleotides are present among the aligned taxa. This provides evidence for how the consensus sequence was generated and helps identify conserved and variable regions within the COI gene.

The consensus sequence represents the most common nucleotide at each position in the multiple sequence alignment of the seven taxa. Variable sites contain phylogenetic information because they represent genetic differences that have accumulated throughout the entire evolution process. Species that share similar nucleotide substitutions are generally more closely related and are more likely to have inherited these changes from a common ancestor. Therefore, variable sites are useful for inferring evolutionary relationships among species.

Phylogenetic Tree

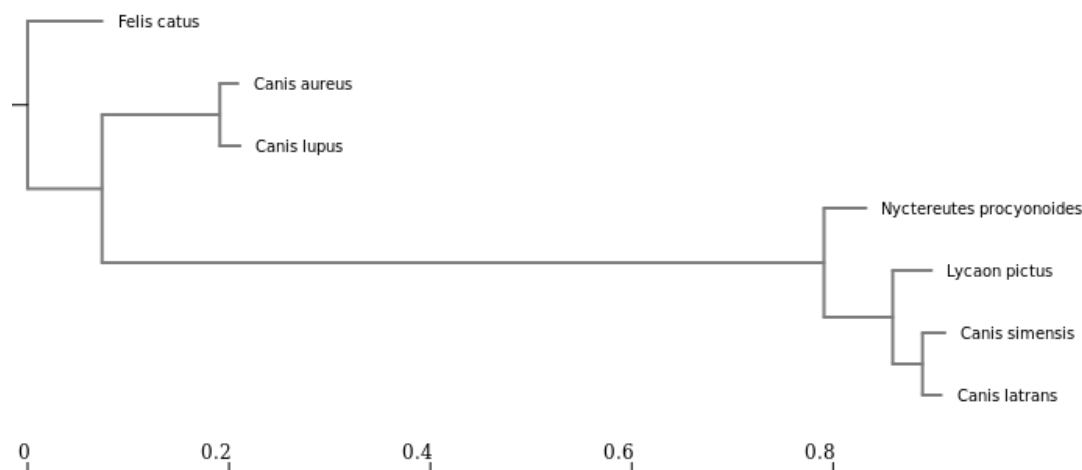


Figure 4. This phylogenetic tree identifies the *Felis catus* as the outgroup because it belongs to the family Felidae, providing a reference point for rooting the tree and distinguishing it from the ingroup. All canid species form a single clade, indicating that they share a common evolutionary ancestor (Senn et al 2022). Within this clade, *Canis aureus* and *Canis lupus* form one pair of sister taxa, while *Canis simensis* and *Canis latrans* form another, indicating that each pair shares a more recent common ancestor compared to the other species. *Lycaon pictus* is sister taxa to the *Canis simensis* and *Canis latrans* clade, whereas *Nyctereutes procyonoides* diverges early within the same lineage. The branch lengths represent the amount of genetic change inferred from the

COI/COX1 gene sequences – the shorter the branches between the taxa indicate lower genetic divergence and closer evolutionary relationships (Liang Liu 2015), while the longer internal branch separating *Canis aureus* and *Canis lupus* lineage from the remaining canids suggest greater genetic divergence and an earlier evolutionary split.

Peer-reviewed Published Accounts

Fuentes-González and Muñoz-Durán (2017) studied the evolutionary relationships of 35 living species of the family Canidae using molecular, morphological, ecological, and other biological data. Their study showed that the relationships among canids can be understood by comparing similarities and differences among species. They also examined the positions of *Nyctereutes* and *Lycaon*, which are included in the present study. Their findings emphasize that canid species may exhibit different evolutionary relationships despite sharing common morphological characteristics. This is important in constructing a phylogenetic tree because molecular data can provide additional evidence for determining how species are related to one another. In the present study, the position of *Nyctereutes procyonoides* as an outgroup and the separation of *Lycaon pictus* from the *Canis* species can likewise be interpreted in relation to their evolutionary divergence. These findings support the use of DNA sequence data to examine the relationships and diversity of the selected Canidae species.

Similarly, Zrzavý et al. (2018) examined the relationships of living and extinct canids using different types of data, including mitochondrial and nuclear genetic markers. Their results supported the grouping of dog-like canids and showed that *Nyctereutes* and *Lycaon* have distinct positions within the Canidae family. The use of both mitochondrial and nuclear genetic information in their study demonstrates how molecular characters can reveal evolutionary relationships that may not always be evident from external morphology alone. Their findings are particularly relevant to the present analysis because the phylogenetic tree also uses DNA sequence information to compare selected canid species. The distinct placement of *Nyctereutes* and *Lycaon* further demonstrates that members of the same family can occupy different evolutionary positions, reflecting their separate evolutionary histories. This study therefore provides additional support for using genetic data to explain similarities, differences, and

evolutionary relationships among canid species, which is also observed in the phylogenetic tree produced in the present study.

Discussion

There is a broad consensus among researchers that COI is the most effective single marker for animal species identification. COI consistently distinguishes closely related species across all major animal phyla, including insects, crustaceans, arachnids, mollusks, fish, amphibians, reptiles, birds, and mammals, with a level of resolution that no other single gene has matched. It was also discussed that researchers can generate new COI sequences using standardized protocols and directly compare them without the need to design custom primers or optimize conditions for each new taxonomic group. Because COI sequences are relatively short and contain high interspecific variability, BLAST searches against NCBI's nucleotides collection are computationally fast and typically return unambiguous top hits with high percent identity and coverage, making it more accessible, faster, reliable, and efficient than sequencing longer genes, multiple panel, or whole mitochondria genomes. Despite its widespread use, COI does not really work well across broad taxonomics especially in plants, fungi, and many protists since these organisms do not possess primer binding sites or show enough variation in COI to be useful—instead you will use a different marker. Another issue is that NCBI itself is not perfectly curated, there are plenty of sequences in the database that are misidentified or lacking proper voucher information. These are true for understudied groups where few verified reference sequences exist. Furthermore, COI is a single mitochondrial gene, which means it tells you about maternal lineage but nothing about hybridization, recent intergression, or population-level processes. The mitochondrial COI gene has several limitations and was encountered by the researchers as conducting the activity. First, the COI struggles to resolve deeper evolutionary relationships within the family, though it can reliably separate clades for both disciplines. Relying also on one single gene means the analysis has limited phylogenetic signal and is vulnerable to artifacts like long-branch attraction, especially when the outgroup is as distantly related as the domestic cat.

Figure 5. Uploaded COI FASTA sequences in the Galaxy History panel.

[illegible]

COI FASTA sequences were combined into a single multi-FASTA dataset using the

COI FASTA sequences were combined into a single multi-FASTA dataset using the

Concatenate dataset tool in Galaxy. The resulting dataset has seven sequences (6 *Canidae* species; 1 outgroup), which was used for the multiple sequence alignment.

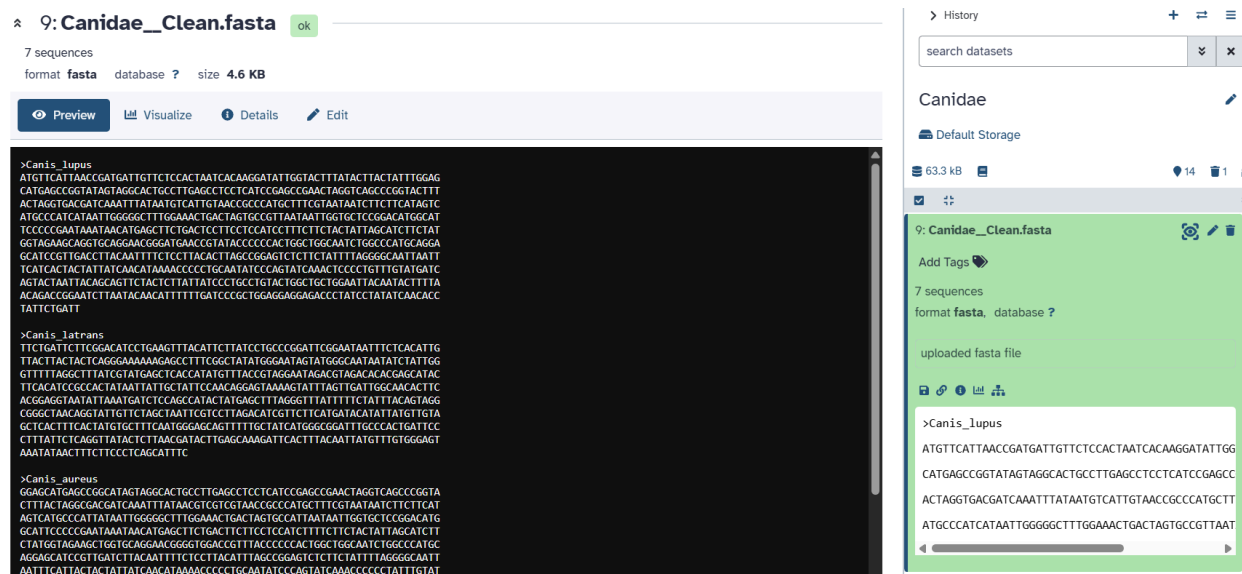


Figure 7. Cleaned multi-FASTA dataset for sequence alignment.

The cleaned version of the multi-FASTA dataset, in which the accession numbers and other unnecessary information in the FASTA headers were removed. This was used for the multiple sequence alignment. Only the species name was retained to ensure that the labels in the resulting phylogenetic tree were clear and easier to distinguish during interpretation.

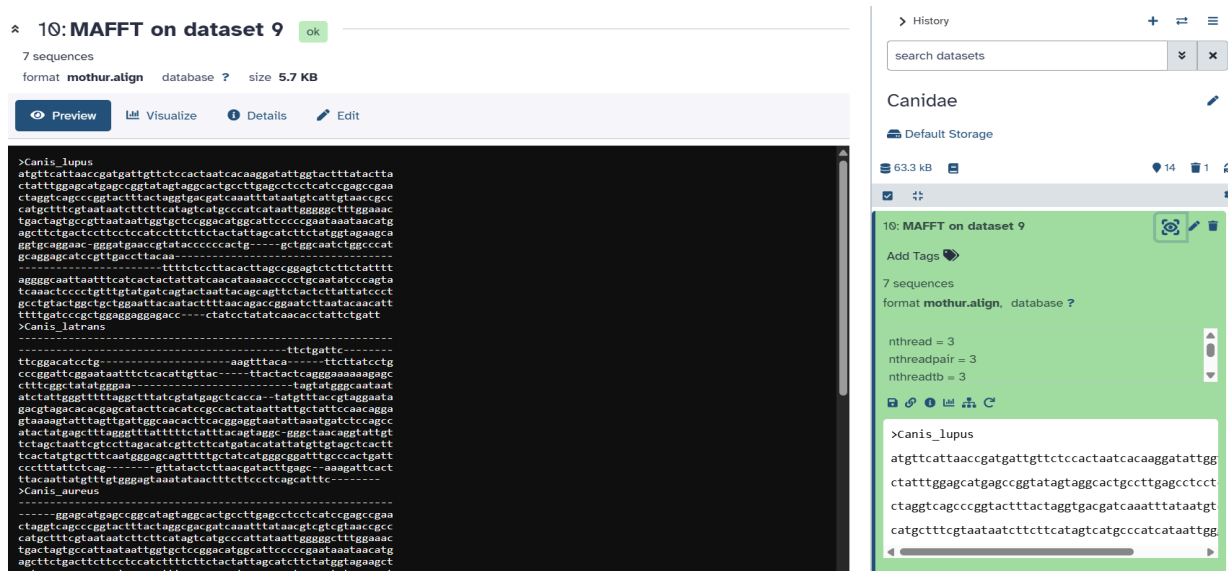


Figure 8. Multiple sequence alignment of COI sequences using MAFFT.

The cleaned multi-FASTA dataset was aligned using MAFFT multiple sequence alignment tool in Galaxy. MAFFT was used because the MUSCLE tool was not available in the Galaxy used for this analysis. The alignment arranged homologous nucleotide positions into columns, while gaps (hyphens) were introduced on parts where it is needed to improve sequence alignment.



Figure 9. Consensus sequence and Consensus summary

The consensus sequence and consensus summary were generated from aligned COI sequences using the Consensus.seqs tool in Galaxy. This image represents the most common nucleotide at each position in the alignment, while consensus summary shows the frequency of each nucleotide, gaps, as well as the base for every alignment.

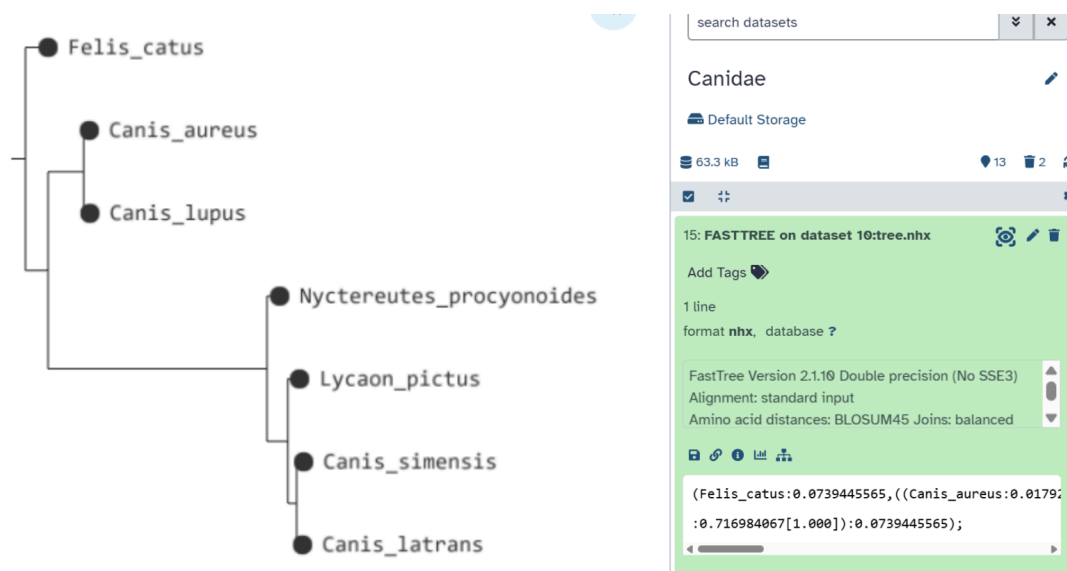


Figure 10. Construction of the phylogenetic tree using FastTree in Galaxy.

The aligned COI sequence was used as the input for the FastTree tool in Galaxy to construct the phylogenetic tree. The results were generated in Newick (nhx) format and then visualized to examine the evolutionary relationships among the selected *Canidae* species and the outgroup (*Felis catus*).

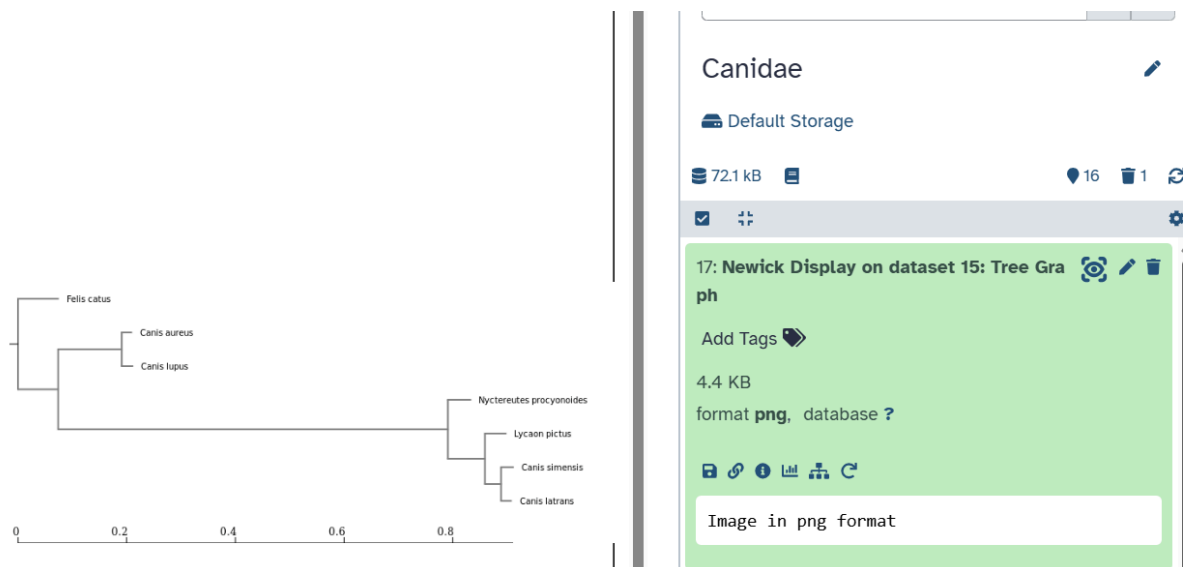


Figure 11. Visualization of phylogenetic tree and branch lengths using Newick Display tool.

The phylogenetic tree generated by FastTree was visualized using the Newick Display tool in Galaxy. This visualization was used to examine the evolutionary relationships of *Canidae* species between one another with *Felis catus* included as an outgroup. The Newick display also shows the scale bar to represent the branch lengths indicating the estimated amount of genetic change among the taxa. This was used because the default FastTree output does not display the branch lengths.

Appendix B. Complete Consensus Summary of the Aligned COI Sequence (Google Sheets)

Due to its length, only a portion of the consensus summary is presented in the main text. The complete consensus summary is provided as a Google Sheet attachment and is available at the following link: [Canidae_Consensus_Summary](#)

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